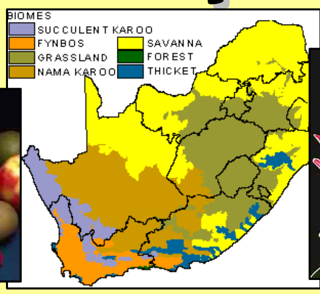
NQF Level: **1** US No: **116158**

Learner Guide

Primary Agriculture

Apply Basic Agricultural Enterprise Selection Principles



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My Workplace: . . . .

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agriculture
Department: Agriculture
REPUBLIC OF SOUTH AFRICA

AgriSETA

Before we start…

Dear Learner,

This Learner Guide contains all the information required to acquire all the knowledge and skills leading to the unit standard:

Title:	Apply basic agricultural enterprise selection principles				
US No:	**116158**	**NQF Level:**	1	**Credits:**	2

The full unit standard is attached at the end of this Learning Guide. Please read the unit standard at your own time. Whilst reading the unit standard, make a note of your questions and aspects that you do not understand, and discuss it with your facilitator.

This unit standard is one of the building blocks in the qualifications listed below. Please mark the qualification you are currently doing:

Title	ID Number	NQF Level	Credits	Mark
National Certificate in Animal Production	48970	1	120	
National Certificate in Mixed Farming Systems	48971	1	120	
National Certificate in Pant Production	48972	1	120	

Please mark the learning program you are enrolled in:

Your facilitator should explain the above concepts to you.

Are you enrolled in a:	Yes	No
Learnership?		
Skills Program?		
Short Course?		

You will also be handed a Learner Workbook. This Learner Workbook should be used in conjunction with this Learner Guide. The Learner Workbook contains the activities that you will be expected to do during the course of your study. Please keep the activities that you have completed as part of your Portfolio of Evidence, which will be required during your final assessment.

You will be assessed during the course of your study. This is called *formative assessment*. You will also be assessed on completion of this unit standard. This is called *summative assessment*. Before your assessment, your assessor will discuss the unit standard with you.

Enjoy this learning experience!

How to use this guide ...

Throughout this guide, you will come across certain re-occurring "boxes". These boxes each represent a certain aspect of the learning process, containing information, which would help you with the identification and understanding of these aspects. The following is a list of these boxes and what they represent:

What does it mean? Each learning field is characterized by unique terms and **definitions** – it is important to know and use these terms and definitions correctly. These terms and definitions are highlighted throughout the guide in this manner.

You will be requested to complete **activities**, which could be group activities, or individual activities. Please remember to complete the activities, as the facilitator will assess it and these will become part of your portfolio of evidence. Activities, whether group or individual activities, will be described in this box.

Examples of certain concepts or principles to help you contextualise them easier, will be shown in this box.

This box indicates a **summary** of concepts that we have covered, and offers you an opportunity to **evaluate** your **own progress** and / or to **ask questions** to your facilitator if you are still feeling unsure of the concepts listed.

My Notes ...

You can use this box to jot down questions you might have, words that you do not understand, instructions given by the facilitator or explanations given by the facilitator or any other remarks that will help you to understand the work better.

What are we going to learn?

No prior learning assumed to be in place for this unit standard

What will I be able to do?

When you have achieved this unit standard, you will:

- Have knowledge about the basic principles of enterprise selection in Agri-business and will be well positioned to extend your learning and practice into crop production and animal production systems.
- Be equipped with adequate skills to have input into enterprise selection and production to improve productivity and performance.
- Recognise and understand the importance of the application of business principles in agricultural production with specific reference to enterprise planning.
- Operate a farming practice as a business and will have gained the knowledge and skills to move from a subsistence orientation to an economic orientation in agriculture, thereby allowing you to gain the knowledge and skills to access mainstream agriculture through a business-orientated approach to agriculture.

What will I know? (Learning Outcomes)

When you have achieved this unit standard, you will have a basic knowledge and understanding of:

- The role of natural resources as well as the role of infrastructure.
- The role and importance of stock as well as of production cycles.
- The role and importance of harvesting as well as post harvest practices.
- Basic farm practices and management skills.

Selecting an Enterprise

Farming is an example of a primary industry. Like a factory, a farm can be seen as a system with a series of inputs, processes and outputs. Processes are the things that go on within the farm. This includes harvesting, ploughing, rearing animals and milking.

The type of farm determines farming systems. Farms can be classified as being arable, pastoral, mixed and market gardening. Different crops grow better in different types of soil. It is therefore very important to identify which soil type is most suitable for your crop type. In South Africa we have different classifications of soil types.

Soil potential is the potential capacity of a specific piece of soil or land to support a specific type of crop in order to get the best quality and quantity produce from that type of crop. Topography is the natural factors that will influence a specific area situated in a specific place, such as the geography of the land.

In this module we are going to consider all the natural resources, infrastructural and structural considerations that you need to take into account when deciding on a type of product for you agri-business. It is also important to consider the variety of crops / products that are related to agriculture and how appropriate they might be to consider.

Session 1 Natural Resources Required

After completing this session, you will be able to:
SO 1: Name natural resources required for the selection of the relevant enterprise.

In South Africa we get **two different types** of farming:

- **Commercial** farming, and
- **Subsistence** farming.

What is the difference?

	Commercial Farm	Subsistence Farm
Formal Definition	A farm which is deemed to be a viable farm operation and which normally produces sufficient income to support a farm family, employ other people and make a financial profit.	A mode of agriculture in which a plot of land produces only enough food to feed the family working it.
Reasons why we might operate each type	Suitable natural resources available.	Minimal natural resources available.
	Good Infrastructure surrounding the area where the farm is situated.	Poor infrastructure around the area where the farm is situated.
	Sufficient capital and cash flow to see the farmer through a production cycle and allow him to produce a crop at a profit.	Lack of capital or sufficient cash flow to allow for more than a short period in which to produce a crop.
Supply of job opportunities & contribution to economy	Normally creates jobs and will offer a contribution to the economy through export.	Does not normally allow for additional employees i.e. no job creation and they do not contribute to the economy except on survival level.

1.1 Farming Systems

Farming is an example of a primary industry. Like a factory, a farm can be seen as a system with a series of **inputs**, **processes** and **outputs**.

Inputs can be divided into **human** and **physical** factors:

- **Human inputs** include labour, capital (money), machinery, seeds, fertiliser and young stock.
- **Physical inputs** include climate and weather, soil, relief (shape of the land) and slope.
- **Processes** are the things that go on within the farm. This includes harvesting, ploughing, rearing animals and milking.

1.2 Classification of Farms

The type of farm determines the farming system. Farms can be classified as being arable, pasture, mixed and market gardening.

- **Arable** farms grow crops.
- **Pasture** farms specialise in rearing animals fed in fields.
- **Mixed** farms consist of both pasture and arable farming.

Farms that have a **high** level of **inputs** are **intensive**. These achieve a high yield per hectare. An example would be a table grape farm in the de Doorns area of the Western Cape.

Those farms that have **low input** and **output** per hectare are **extensive**. An example would be a sheep farm in the Karoo.

Farms can also be classified by what happens to their outputs.

On **subsistence** farms the farmer consumes the produce. Any surplus is usually sold to buy other goods.

- Farms that sell the majority of their produce are known as **commercial** farms.

What is the difference between Extensive and Intensive Farming?

Definition

	Extensive	**Intensive**
Commercial	Sheep farming in Karoo. The poor soils and harsh climate make this area ideal for sheep farming.	Market gardening in the North West Province. Good Soils and a mild climate make for good conditions.
Subsistence	Nomadic pasturalism in Eastern Cape	Piggery in areas of the Western Cape.

1.3 Requirements for Commercial Farming

In order to become a successful commercial farmer, the following resources need to be in place:

- ♦ Natural resources.
- ♦ Human resources.
- ♦ Infrastructure.
- ♦ Capital.
- ♦ Crops and structures.

The **natural resources** required when selecting a relevant enterprise includes:

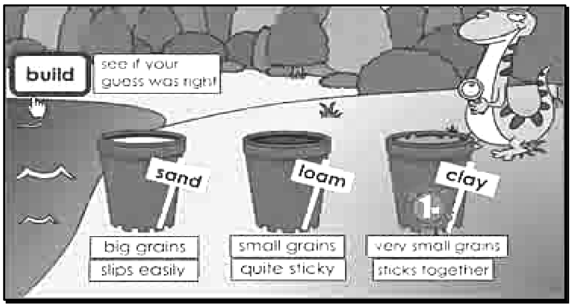
- ♦ A suitable climate;
- ♦ Suitable soil properties; and
- ♦ Sufficient water of suitable quality.

Soils to Grow a Bumper Crop

Different crops are better suited to different soil types. It is important to identify the soil type and soil characteristics, most suited to the crop you want to produce.

South African soils are classified according to different characteristics.
The major soil types are Sand, Loam and clay soils. Within these broad groupings further divisions are made as set out below.

Textural Group		**Textural Classes**
Sandy	Coarse	Sand Loamy Sand
Loamy	Moderately Course	Sandy Loam
	Medium	Loam Silt Loam Silt
	Moderately Fine	Clay Loam Sandy Clay Loam Silty Clay Loam
Clay	Fine	Sandy Clay Silty Clay Clay



Loam. When rubbed between the thumb and fingers, approximately equal influence of sand, silt, and clay is felt. Makes a weak ribbon (less than 2.5 cm long).

Sandy loam. Varies from very fine loam to very coarse. Feels quite sandy or gritty, but contains some silt and a small amount of clay. The amount of silt and clay is sufficient to hold the soil together when moist. Makes a weak ribbon (less than 2.5 cm long).

Silt loam. Silt is the dominant particle in silt loam, which feels quite smooth or floury when rubbed between the thumb and fingers. Makes a weak ribbon (less than 2.5 cm long).

Silty clay loam. Noticeable amounts of both silt and clay are present. Makes a medium ribbon (2.5 to 5 cm long).

Clay loam. Clay dominates a clay loam, which is smooth when dry and slick/sticky when wet. Silt and sand are usually present in noticeable amounts in this texture of soil, but are overshadowed by clay. Makes a medium ribbon (2.5 to 5 cm long).

Soil Potential

Soil potential is the potential capacity of a specific field to support a specific crop in order to obtain the highest yields of top quality produce expected of the crop.

ACTIVITY

Please complete activity **1.1** in your workbook

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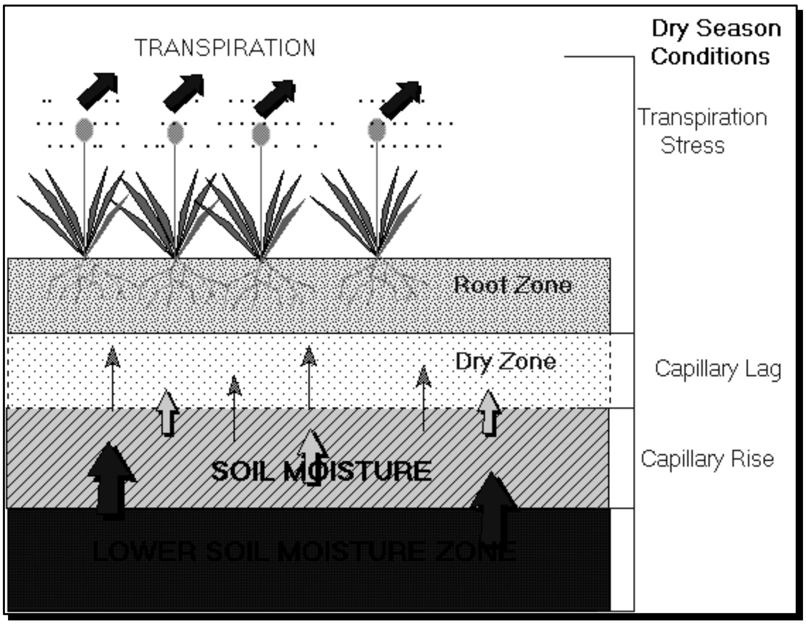
Water Sources

Water sources include municipal piped water, bore holes, ponds, reservoirs, canals, ditches, streams and rivers. The water used in farming must be of a quality suitable to its use. Water used for micro irrigation must be clean of particles that could block the small holes of dripper or micro jet irrigation devices.



- Access to water is crucial to success of any farm. To save on installation costs the use of gravity fed systems should be exploited to its maximum and the pumps kept to a minimum.
- Rather use the wastewater to irrigate and fertilise fodder crop.
- **Ground water** accessed through boreholes is generally of good quality. The quality of water used needs to be checked before the enterprise is initiated. Water quality should be suitable for the production system that is to be used.
- **Surface water** such as streams, springs and dams can be used, but you should guard against bacteria, algae and other aquatic life.

The availability and quality of water sources will determine to a large extent the type of farming to be practiced. Two of the cropping systems that are highly dependant on water are irrigation farming and hydroponics.

Irrigation Farming

Irrigation is a key factor in the agricultural economy of many parts of the world.

Irrigation farming refers to a system where water is moved from a water source to the area where farming activities are taking place trough furrows, pipes or other systems. The water is then applied manually or mechanically to the farming activity by means of sprinklers, dripping etc.

Hydroponics Farming

Hydroponics farming is the growing of plants in a water medium without soil.

Nutrients are added to the water making it directly available to the plant.

The availability of water sources is not only important to crop growers, nut also to livestock farmers. Water is required as drinking water for livestock, whether free roaming or penned. Drinking water quality must be good, similar to the quality of drinking water for humans. Water is also used in livestock farming to clean the areas where the stock is housed. This includes cleaning of chicken pens and pigsties. It is important that the wastewater generated during cleaning is properly treated and that it does not contaminate the sources of clean water.

Another important consideration is run-off wastewater, especially where it may be carrying high levels of nitrogen such as in the case of pig and dairy farming. This nutrient rich water should not be allowed to find its way back into the natural watercourse as this will lead to eutrophication.

ACTIVITY

Please complete activity **1.2** in your workbook

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1.4 Vegetation

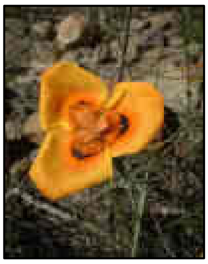
The richness in diversity of the South African flora is acknowledged worldwide. However there are two types of vegetation that you should be aware of in terms of agriculture:

Indigenous Vegetation

This is defined as trees, under storey plants, groundcover and plants occurring in a **specific area** only. Examples of such types of Vegetation for South Africa are Proteas, Fynbos, Rhino Veld, Disas and many more.

Protea

Fynbos



Rhino Veld

Disas

Invasive Vegetation

These are alien plants and plants that have been brought to South Africa from other countries for their beauty, economic value or ecological purpose. Some are brought in unintentionally and here, without their natural enemies, are able to reproduce and spread prolifically.

The plants or seeds enter the country in a number of different ways: for example on people's shoes, tents, by mail order on ships and planes. Even animals that cross the borders can bring seeds in. The invader plants and seeds spread rapidly and take up the growing space of our own indigenous plants.

Invasive alien plants threaten the indigenous vegetation as they use up valuable and limited water resources. Most of them drink more water than indigenous plants and are depleting the valuable underground water resources. Many invasive plants are also responsible for causing exceptionally hot fires and affect the makeup of soil structure.

Fields covered in invasive vegetation will have to be cleared before the area can be farmed. This could be a costly exercise.

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Invasive species can be classified into 3 categories as indicated in the table below:

Category	Description	Examples
1: Declared weeds	These are plants that must be controlled (removed) on land or water surfaces by all land users. These plants may no longer be planted or propagated and all trade in their seeds, cuttings or any other propagation material is prohibited.	Lantana - Pom pom weed - Bugweed - Azolla - Queen of the night - Pampas grass – Cat's claw creeper - Red sesbania - Yellow oleander - Yellow bells - Water hyacinth.
2: Declared Invaders (plants with *commercial* value)	These are invader plants that pose a threat to the environment but nevertheless can be exploited for timber, fruits, fuel wood, medicinal plants, animal fodder, building material or shelter of to stabilise soil. These species are only allowed to occur in demarcated areas. If the plants are used for commercial purposes, land users have to obtain a water use licence as these plants consume large volumes of water. Where plants occur outside demarcated areas they have to be removed.	Black wattle - Patula pine - Sisal - Red eye - Grey poplar - Watercress - Port Jackson willow - Guava - Cluster pine - Honey locust - Weeping willow (not to be confused with the indigenous willow).
3: Declared Invaders (plants with *ornamental* value)	These are plants that have the potential of becoming invasive but are considered to have ornamental value. In terms of Regulation 15 of CARA, these plants will not be allowed to occur anywhere except in biological control reserves unless they were already in existence when these regulations came into effect (30 March 2001). This means that existing plants do not have to be removed by the land user, however they must be kept under control and no new planting may be initiated and the plants may no longer be sold.	Jacaranda - Syringa - Australian silky oak - St Joseph's lily - Sword fern - Tipu tree - New Zealand Christmas tree.

South African Veld Types

South African Veld types can be broken up into about 70 different groups based on its production potential and ecological capacity.

In other words, a veld type is a vegetation unit with the same ranching potential through out it's range. These 70 veld types can be reduced to 13 based on habitat preferences and game distribution, namely Fynbos, desert, & succulent Karoo, arid Karoo, great Karoo, forest with grass veld, valley bushveld, Lowveld, sweet bush veld, Northern Province sour and sourish mixed bushveld, Kalahari, short & mixed grass veld, mountain grass veld and tall grass veld.

However, veld types can, from a grazing point of view, be broken down into 3 groups namely sweet, sour and mixed. (See table on next page.)

South African Veld Types:

Veld Type	Description
Sweet	• Found in the lower-laying, frost-free parts of the country. • Are palatable throughout the year. • Has a high nutritional value. • Keeps animals in good condition and reproductive state. • Associated with lower rainfall (250mm – 500mm per annum).
Sour	• Carries grazing vegetation that loses its palatability and nutritional value at maturity, therefore only valuable during the growing season. • Are found in rainfall areas of 625mm and above.
Mixed	• Occurs in the transitional zones between sweat & sour. • The soil pH is lower than in the sweet veldt types resulting in lower calcium and phosphate quality in the soil.

1.5 Climatic Conditions

The layout and activities of a farm will be influenced by **regional climatic conditions** and **micro climatic conditions** found in the area. Various crops are adapted to specific environmental conditions. Similarly livestock may have adapted of are bred to survive under specific environmental conditions. Before a farming systems are initiated, the climatic conditions of the area must be determined. The suitability of specific crop of livestock types must be established before you start your enterprise. Under certain conditions, such as hydroponics in climate-controlled tunnels, the environmental conditions of the location of the farm may be irrelevant. This is because the farmer will artificially control the climatic conditions. The section below is a summary of the **major climatic areas of South Africa:**

- ♦ In South Africa, summer rainfall areas include the Northern, Southern and Western Cape as well as the East coast and Highveld.
- ♦ The South African West coast and interior experiences dry conditions due mainly to the cold Benguela current running South North in the Atlantic Ocean.
- ♦ The South & South Eastern coastlines experience increasing wetter conditions as you move North East due to the warm Agulhus current. Long and high mountain ranges running parallel to the coastlines result in inland rain shadows and drier conditions.

ACTIVITY

Please complete activity **1.3** in your workbook

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1.6 Effect of Topography on Enterprise Selection

Topography

Topography describes an area and the natural factors that influence a specific area situated in a specific place, such as:

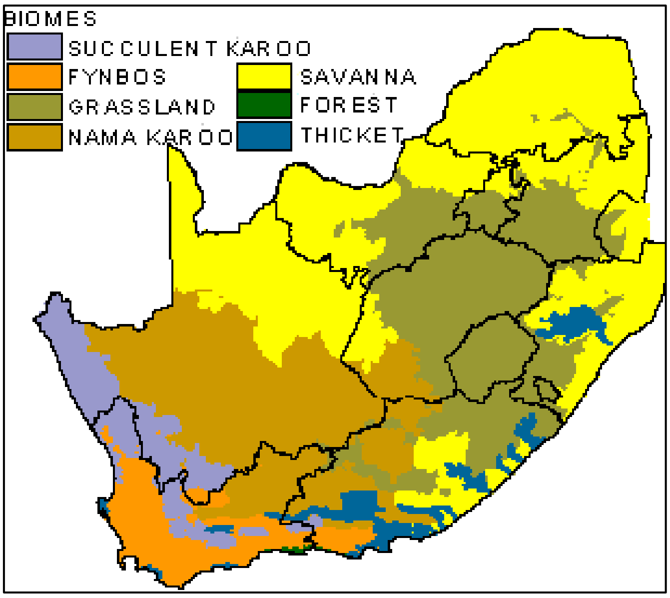
- The shape and lie of the land (geography);
- Its relation to oceans and mountain ranges;
- Its climate;
- Its geographical history;
- Its slope directions and elevation; and
- Its location in terms of latitude and longitude.

Topography of South Africa

South Africa lies between 22 and 35 degrees south, flanked on the west by the Atlantic Ocean and on the east by the Indian Ocean, whose waters meet at the country's most southern tip, Cape Agulhas. The long coastline stretches more than

2 500 kilometres from a desert border in the north-west, down the icy and treacherous Skeleton Coast to Cape Agulhas, then up along rolling green hills and wide beaches fronting the warm Indian Ocean, to a border with subtropical Mozambique in the north east.

1.7 Geography and Climate of South Africa



South Africa occupies the southern tip of Africa, its long coastline stretching more than 2 500km from the desert border with Namibia on the Atlantic coast southwards around the tip of Africa and then north to the border with subtropical Mozambique on the Indian Ocean.

The low-lying coastal zone is narrow for much of that distance, soon giving way to a mountainous escarpment that separates it from the high inland plateau. In some area such as KwaZulu-Natal, a greater distance separates the coast from the escarpment.

Although the country is classified as semi-arid, it has considerable **variation** in **climate** as well as **topography.** These factors need to be taken into account when selecting an agricultural enterprise. Ensure that the area is suitable for the enterprise you want to develop. The section below briefly discusses the different regions found in South Africa. You will have to determine for yourself which enterprises are suited to the different areas.

Karoo Plateau

The **great inland Karoo plateau**, consists of sparsely populated scrubland, is very dry. The north-west areas towards the Kalahari desert are the driest. It is obvious that these areas will not be suited to farming of crops like vegetables, especially where not water is available for irrigation. The area becomes extremely hot in summer and can be icy cold in winter.

Eastern Coastline

The **eastern coastline** is a lush and well-watered area, where frost is not frequently experienced. The southern coast (part of which is known as the Garden Route) is less tropical but also green, as is the Cape of Good Hope. This south-western corner of the country has a Mediterranean climate, with winters rainfall and hot, dry summers. Its most famous climatic characteristic is its wind, which blows intermittently virtually all year round, either from the south-east or the north-west.

Average temperatures in ºC		
	Summer	**Winter**
Cape Town	20	12.6
Durban	23.6	17
Johannesburg	19.4	11.1
Pretoria	22.4	12.9

Source: Lew Leppan: The South African Book of Records. Cape Town, Don Nelson, 1999.

Highveld

The highveld is found north of the Vaal River and receive more rain than the Karoo region and free state. The area is at a high altitude, Johannesburg being at 1 740m. Winters are cold, but snow is rare.

Lowveld

The Lowveld is situated to the north and to the east of the country. Temperatures are higher than the highveld.

Climate versus Crop

Specific crops need specific climatic conditions in order to produce a maximum quality and quantity crop:

♦ Deciduous Fruit, Stone Fruit, Pome Fruit & Wine Grapes

Cultivars differ with respect to the intensity of cold required for them to drop their leaves and then to begin budding again simultaneously. Some apple cultivars need more sunshine to develop good colour. Cold, wind, rain, sunshine and humidity together constitute the climate, which determines the growth vigour and the crop quality, and size produced by the fruit trees.

♦ Subtropical Fruit

Different cultivars are tolerant to a wide range of climatic conditions. It is successfully cultivated, under ideal conditions, which vary from very hot, very humid to cool and dry, to very hot and arid. Climatic conditions in a specific area will, firstly determine whether mangoes can be cultivated commercially in the given area, and secondly, will influence the choice of cultivars.

It is advised that the average temperature during winter should preferably be above 5°C. For optimum growth, and production, the average maximum temperature should be between 27 and 36°C.

ACTIVITY

Please complete activity **1.4** in your workbook

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How am I doing?

Concept (SO 1)	I understand this concept	Questions that I still would like to ask
Soils are examined and suitability for cultivation is assessed.		
Water sources are identified.		
Climatic conditions are identified and described.		
Basic vegetation types are identified.		

Session 2 Infrastructure Requirements

After completing this session, you will be able to:
SO 2: Describe infrastructure requirements for the selection of the relevant enterprise.

2.1 Types of Infrastructure

Before you select an area to start the enterprise you need to ensure that the relevant infrastructure has been developed. Infrastructure that may be required include, but are not limited to:

- Electricity.
- Road / rail access for transport.
- Access to water for irrigation.
- Access to communication (either land lines or Cellular)
- Access to suppliers of supplies.
- Access to markets.

If you decide for instance to start an enterprise that grows vegetables under hydroponics conditions, you may only require a small piece of land. The climatic conditions of the area will not be all that important as you will control these in the tunnels. You will however require a source of good quality water, reliable source of electricity and reliable transport system. You will also have to establish structures for growing the crop as well as storage and packing facilities. Access to reliable telecommunications infrastructure is also important.

It is essential that the infrastructure needs of the enterprise you want to develop are known **before** you start the business. You may have a viable business idea, but the area that you target for the development of the enterprise may not be suitable. In most cases banks or institutions offering extension services will be able to guide you in your business.

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Infrastructure vs. Structures

Definition

Infrastructure is a combined name for the things that are the bare essentials that you require in order to be able to operate a farm. Infrastructure is often the responsibility of other people beside yourself, such as ESCOM, TELKOM the Cellular Networks, National Government and Local Government.

Structures are the things that you may decide to build / erect on your farm, yourself, in order to ensure that you may produce optimal crops. These would include things like: Fencing, Irrigation Systems, Trellising systems, Roads on the farm, Stores and Sheds, Water storage dams and tanks.

Activity

Please complete activity **2.1** in your workbook

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2.2 The Role of Infrastructure in Agri-business

Farm infrastructure may include roads, electricity and water supply.

Irrigation systems reed a source of good quality water, reservoirs and a piping system that feed to and from the reservoirs. Packing and milking sheds require electricity and water. Access to fields is important, so having suitable roads available is important. When the farm enterprise is being planned, you should take into account the infrastructure required, as well as its layout.

The layout of infrastructure refers to the positioning of the different pieces of infrastructure. It is important for example where the source of electricity is positioned, and where it is required. Planning the layout of the farm before the development is started, could save you much time, money and effort in the long term.

2.3 Infrastructure Placement

The positioning of infrastructure will be determined by the topography of the farm. If for example the slopes on the farm are steeper than 45 degrees, it would be very difficult to erect trellising systems or to operate tractors on these slopes. Steep slopes will also lead to erosion from run-off water if not managed well.

If you plan on selling you produce say as dried fruit, you will have to have a suitable area to place a drying facility. If the farm is located in an area with high humidity, you will have to consider using electric drying systems. To operate these, you will have to have access to electricity.

ACTIVITY

Please complete activity **2.2** in your workbook

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How am I doing?

Concept (SO 1)	I understand this concept	Questions that I still would like to ask
Types of infrastructure are described.		
The role and function of infrastructure is understood.		
The availability of infrastructure is determined.		
The suitability of landscape for infrastructure is determined.		

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Session 3 Identify Crops and Animals Suitable to your Agri-Business

After completing this session, you should be able to:
SO 3: Identify appropriate crops and/or animals for the relevant enterprise.

3.1 Identifying Appropriate Crops and/or Animals for the Relevant Enterprise

Before you start the enterprise you need to select the crops, the cultivar of variety or type and race of livestock you want to use in the enterprise. Remember the animal of crop is the potential source of end product you wish to sell. So the crop or animal is the source of your profit.

You therefore must ensure that the type you select will provide you potentially with the highest profit margin. It is essential that you select the type that suits your environment and farming system you plan to use. Furthermore you need to select the type that is suited to your environment and has some resistance or immunity to diseases that are potentially found in your area.

ACTIVITY

Please complete activity **3.1** in your workbook

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Farming Systems

The section below provides a brief explanation of different farming systems.

- **Arable** farms grow crops.
- **Pasture** farms specialise in rearing animals.
- **Mixed farms produce** both pastures and crops
- Farms that have a **high** level of **inputs** are **intensive**. These achieve a high yield per hectare. An example would be a table grape farm in the de Doorns area of the Western Cape.
- Those farms that have **low input** and **output** per hectare are **extensive**. An example would be a sheep farm in the Karoo.

- On **subsistence** farms the farmer consumes the produce. Any surplus is usually sold to buy other goods.
- Dry land farming is an agricultural technique for cultivating land, which receives little rainfall and there are no dams or irrigation systems installed on the farm.
- Irrigated Farming - Irrigation is a key factor in the agricultural economy of many parts of the world. Irrigation farming refers to a system where water is moved from a water source to the area where farming activities are taking place trough furrows, pipes or other systems. The water is then applied manually or mechanically to the farming activity by means of sprinklers, dripping etc.
- Hydroponics farming is the growing of plants in a water medium without soil. There are nutrients added to the water. It is the most productive way to grow all varieties of plants, and those raised in a hydroponics system will exhibit maximum yield, flavour, vitamin and essential oil content
- Farms that sell the majority of their produce are known as **commercial** farms.
- **Commercial Farms** are those which are deemed to be a viable farm operation and which normally produces sufficient income to support a farm family, employ other people and make a financial profit.
- **Subsistence Farming is a** mode of agriculture in which a plot of land produces only enough food to feed the family working it

ACTIVITY

Please complete activity **3.2** in your workbook

My Notes ...

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3.2 Crop Characteristics

When considering the characteristics of the crop type of your choice, you need to take into account how plants relate to their environment.

It is important to remember that there are seven (7) requirements for plants to grow: **Space to Grow**; **Temperature**; **Light**; **Water**; **Air**; **Nutrients**; and **Time**.

Space to Grow

All plants need space to grow in. The above ground parts of the plant need space for leaves to expand and absorb light energy. Roots also need space to grow. Plants with limited root growth will have minimal development. These aspects are important to consider, in that soil depth could determine the extent of root growth and thus the development of the plant.

Temperature

Temperature affects the extent of plant growth. In most cases crop plants will grow optimally a day temperature that do not exceed 30°C and night temperature that do not go below 15°C. Some plants are adapted to other temperature regimes though breeding or natural selection. It is important that you know the temperature limitations of the crop, and also the general temperature regimes that may occur in the area where you want to farm. Remember the better the plant grows the better the profit margin is likely to be. So grow the plants under temperature conditions that will lead to optimal growth.

Light

Plants need to be exposed to direct sunlight. Different types and cultivars are adapted to different light intensities and day lengths. Some crops like Hops require a 14 to 16 hour day period for optimal growth. Hops is grown in field and additional lighting provided, lengthening the day period, ensuring optimal yields.

Water

Water is critical for plant to survive. Different plants require different amounts of water for optimal growth. Make sure that you understand the water requirement of the crop. Some crops can only be grown under irrigation in certain parts of the country. One such crop is Pistachio nuts grown on the banks of the Orange River. The area does not receive much rainfall, but water is provided through irrigation from the Orange River. Much infrastructure had to be developed at great expense for the establishment of this crop. Drainage is also an important factor to consider when talking about water – if plants are not in a place with good drainage; they will stand with their roots in water and will die as a result of lack of oxygen.

Air

Plants use carbon dioxide in the air and return oxygen. Smoke, gases, and other air pollutants can damage plants, reducing their productivity.

Nutrients

Most of the nutrients that a plant needs are dissolved in water and then taken up by the plant through its roots. Fertilizers will help to keep the soil supplied with nutrients a plant needs.

Time

It takes time to grow and care for plants. Some plants require more time to grow than others. Getting plants to flower or fruit at a certain time can be challenging. Plants that normally grow outdoors need a certain number of days to flower or fruit. You can time plants to flower or fruit on a certain date. This is a good lesson in both plant science and math.

3.3 Requirements for a Specific Crop

The section deals with plant requirement for growth for fruit trees as an example. Not all crops have the same requirement for optimal growth these requirements must be determined for your crop type before you start your enterprise.

Requirement for Fruit Trees

- **Soil.** Fruit trees need deep soils to grow. They also do not grow well in clay soils as the need good drainage in order to flourish.
- **Water.** Fruit trees need a large amount of water, but you can manipulate its water requirements at different times in the growth cycle. (6 000 000 000 - 7 500 000 000 litres per Ha! for the lifetime of one orchard). Fruit trees cannot survive without any water for a prolonged period of time. At the same time fruit trees are not happy to grow in standing water, and especially peach and apricot trees will easily die if they are in a place without good drainage.
- **Sunlight & Temperature.** Fruit trees need sunlight in order to photosynthesise in order to survive. Fruit trees also need to be in an area with sufficient cold in winter to ensure that they build up enough cold units to regenerate in the next growth season.
- **Air.** Fruit trees need a balanced amount of oxygen and carbon dioxide in order to grow proficiently and produce optimum fruit. If they are in an area where the air is polluted they will not be able to photosynthesise effectively.

How am I doing?

Concept (SO 3)	I understand this concept	Questions that I still would like to ask
Different livestock or crop types are described.		
Characteristics of the different types are explained.		
Different uses of the different types are identified.		
The suitability of infrastructure for livestock or crops is determined.		

My Notes ...

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Session 4 The Production Cycle

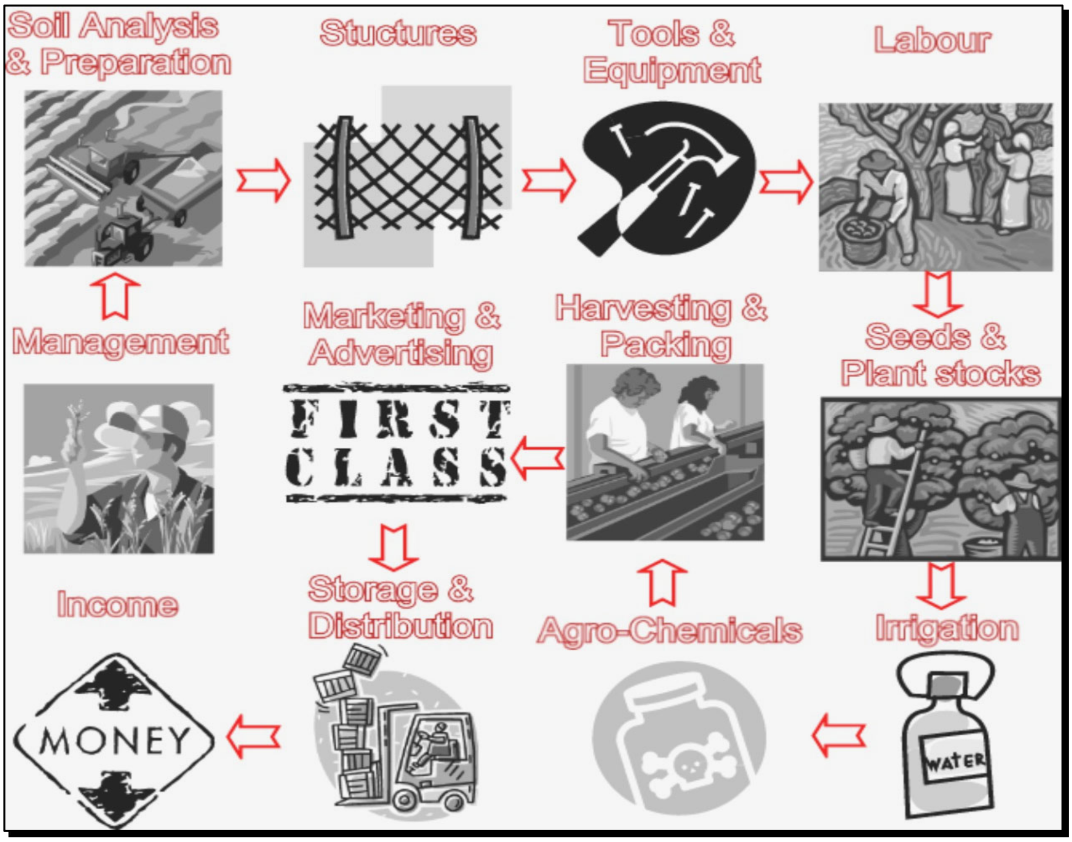
After completing this session, you will be able to:
SO 4: Identify production cycle within relevant enterprise.

4.1 The Production Cycle Within Agri-Business Enterprises

A production cycle is a summary of the major events, processes and parties involved in establishing a farming enterprise and produce a product that the farmer sells to make a profit. Agricultural business will at some stage of the cycle either supply or require the inputs in their business involving:

- Services
- Goods
- Requirements/Needs
- Marketplace

The needs of a farm are centered around maximum profits. The needs of the consumer is for value products at reasonable prices, provided at consistent quality and that is not damaging to their health.



The services that a farm might offer include:

- Transportation of the produce directly the wholesaler.
- On-farm accommodation.

The services that a farm might need include:

- Specialist advice on pesticides.
- Telecommunication.
- Financial institution's assistance.
- Irrigation specialists.

The goods that a farm might offer include:

- The fruit that has been produced.
- A processed version of the fruit that has been produced such as dried fruit or juice.

The goods that a farm might require include:

- Agro-chemicals.
- Vehicles.
- Packaging.
- Parts.
- Office furniture.

The needs that a farm might fulfil include:

- The consumer's demands.
- The consumer's demand for a specific type of produce.
- The consumer's demand for a specific processed product.

The needs that a farm has include:

- Financial.
- Structural.
- Infrastructure.
- Human Resource related.
- Natural resource related.

The route of supply to the market can be:

- Direct to the consumer via farm stalls or direct sales.
- To a packing facility for packing, grading and sorting.
- To a wholesaler if the farm has its own pack house.
- To an export market if the farm has its own pack house.

Certain markets also supply the farm with goods and services that include:

- Agro-chemical.
- Packaging.
- Transportation.

4.2 Characteristics of a Production Cycle

An Agri-business' operating costs can be divided into 6 Groups. Each of these 6 groups has a direct or indirect influence on production, and each of the 6 groups' costs varies or stays the same as production takes place. These 6 groups are:

- Money or Capital
- Management
- Employees
- Plant & Equipment (Tools & Machinery)
- Material to grow / supply and package your agri-product
- Market

Agri-business is normally a seasonal business. It sometimes takes a few years or seasons for a farm or agri-business to produce the product that it set out to produce.

However – there are "inputs" needed through the growth cycle. It is thus very important to plan very carefully – so that you make sure that when you eventually start making a profit that you also recoup all the accumulative expenses that you had previously incurred!

4.3 Comparison of Production Cycles

Production cycle for different crops and live-stock types differ. What follows is an example of a production cycle for fruit harvested from tree crops.

Example

Phase 1:
Starts at the set of the fruit
Lasts 6-10 days

Phase 2:
The seeds develop
Cell division is rapid
Lasts 3-6 weeks

Phase 3:
The fruit enlargement slows down
Lasts 2-4 weeks

Phase 4:
Rapid increase in fruit mass
Accumulation of sugars & Carbohydrates

The diagram provides seasonal the production cycle for a tree crop, once it has been established and the trees bear fruit. Remember that the production cycle for the crop may have started three to five years earlier with the establishment of the orchard. This would have included the selection of the site, the testing of soil for cop suitability etc. Once selected the soil required preparation and the trees were sourced and planted. During the three to five years that have lapsed, the trees had to be treated for diseases and pests etc. Fertilizers and irrigation were applied despite the lack of fruit for harvest.

The production cycle for a normal season for bearing trees would typically include:

- Irrigation and fertilization during vegetative growth
- Pollination of flowers - bees are required, so farmers need to establish or rent in hives.
- Fruit development and fruit growth
- Fruit ripening
- Fruit ready for harvest

During each of the development stages specific activities need to occur for the crop to be a success. These could include pruning, fertilization, pest and disease management etc.

ACTIVITY

Please complete activity **4.1** in your workbook

My Notes ...

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How am I doing?

Concept (SO 4)	I understand this concept well	Questions that I still would like to ask
Characteristics of a production cycle are described.		
The different production cycles are compared.		
The appropriate production cycle is described correctly.		
Implementation of the production cycle is observed and reported on.		

Session 5 Harvest/Post-harvest Practices within an Agri-business

After completing this session, you will be able to:
SO 5 & 6: Identify harvest / post-harvest practice within the relevant enterprise.

This chapter deals with the harvesting of a crop or product that is produced in the agri-business. The specifications and procedures for different crops and products differ. The adherence to product specific procedures and quality guidelines will however influence the quality of the product being produce.
The quality will determine the acceptability of the product to the market, which in turn will dictate the price that you receive for the product.

This in turn determines the profit you make and the long-term success of your agri-business. It is therefore critical that you identify and understand the harvest and post-harvest procedures for your product, **before** harvest is initiated. The sections below provide an overview of some of the concepts of harvest and post harvest handling.

5.1 Harvest Practices in Agriculture

Harvesting is the process of gathering mature crops from the fields. The harvest marks the end of the growing season, or the end of the crop production cycle for a crop.

Harvesting also encompasses the post-harvest handling of produce, including all the actions required from physically removing the produce, sorting, cleaning and packing. It included both storing and shipping to the market.

Harvest timing critical decision. Often the time of harvest must be selected based on a balance of possible adverse weather conditions and the degree of crop maturity. Occurrences such as frost, and unseasonably warm or cold periods affect the yield and quality of the crop. The extent to which a crop may be influenced is dependant on the crop, and may differ between crops.

An earlier harvest date may avoid damaging conditions, but could result in poorer yield and quality. Delaying harvest may allow for a better harvest, but increases the chance of weather problems. Timing of the harvest often involves a significant degree of risk and gambling.

On smaller farms with minimal mechanization, harvesting is the most labour-intensive activity of the growing season. On large, mechanized farms, harvesting utilizes the most expensive and sophisticated farm machinery, like the combine harvester.

Harvest commonly refers to grain and produce, but is used in reference to fish and timber.

5.2 Post-harvest Handling

Post-harvest handling is the stage of crop production immediately following harvest, including cooling, cleaning, sorting and packing. The instant a crop is removed from the ground, or separated from its parent plant, it begins to deteriorate. Post-harvest treatment largely determines final quality, whether a crop is sold for fresh consumption, or used as an ingredient in a processed food product.

The most important goals of post-harvest handling are keeping the product cool, to avoid moisture loss and slow down undesirable chemical changes, and avoiding physical damage such as bruising, to delay spoilage. Sanitation is also an important factor, to reduce the possibility of pathogens that could be carried by fresh produce, for example, as residue from contaminated washing water.

After the field, post-harvest processing is usually continued in a packhouse. This can be a simple shed, providing shade and running water, or a large-scale, sophisticated, mechanized facility, with conveyor belts, automated sorting and packing stations, walk-in coolers and the like. In mechanized harvesting, processing may also begin as part of the actual harvest process, with initial cleaning and sorting performed by the harvesting machinery.

Initial post-harvest storage conditions are critical to maintaining quality. Each crop has an optimum range for storage temperature and humidity. Also, certain crops cannot be effectively stored together, as they can have unfavourable chemical interactions.

Various methods of high-speed cooling, and sophisticated refrigerated and atmosphere-controlled environments, are employed to prolong freshness, particularly in large-scale operations.

Regardless of the scale of harvest, from home garden to industrialized farm, the basic principles of post-harvest handling for most crops are the same:

- **Handle with care to avoid damage** (cutting, crushing, bruising)
- **Cool immediately and maintain in cool conditions**
- **Cull** (remove damaged items)

5.3 Characteristics of Harvesting and Post-harvest Practices

- **Step 1:** Budget, plan and forecast.
- **Step 2:** Don't harvest until the crop is ready.
- **Step 3:** Adhere to withholding periods for agro-chemical applications.
- **Step 4:** Adhere to specific instructions during harvest – i.e. prevention of bruising.
- **Step 5:** Select your crop for quality and grade.
- **Step 6:** Maintain health and safety of workers during harvest.
- **Step 7:** Plan and manage the harvest to maintain optimum food safety.
- **Step 8:** Plan and manage the harvest to maintain optimum crop quality.
- **Step 9:** Manage costs, equipment and staff to maximise profits.

5.4 The Importance of Health and Hygiene during Harvest and Post-harvest

Personal protective clothing that should be worn?

HAIRNET: Should cover the surface area of your head with no exposed hair follicles. Must be worn by all employees.

BEARD NET: Should be used if you wear beard.

OVERALL/COAT: Should be of the correct size. Must be "zipped up" at all times and the sleeves should not be folded up.

OVERALL/PANTS: Should not be short or torn.

SAFETY BOOTS: Must be worn at all times in a factory. It must be laced-up and a pair of socks must be worn.

5.5 The Importance of Maintaining Quality of your Crop during Harvest and Post-harvest

Regardless of the scale of harvest, from home garden to industrialized farm, the basic principles of post-harvest handling for most crops are the same:

- Handle with **care** to avoid damage (cutting, crushing, bruising).
- Cool immediately and maintain under **cool** conditions.
- **Cull** (remove and dispose of damaged items).

ACTIVITY

Please complete activity **5.1** in your workbook

My Notes ...

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How am I doing?

Concept (SO 5)	I understand this concept well	Questions that I still would like to ask
A characteristic of harvesting practices is described.		
Harvesting practices are understood.		
Importance of health and hygiene is understood.		
Importance of quality is understood.		
Characteristics of post harvest practices are described.		
Post-harvesting practices are understood.		
Importance of health and hygiene is understood.		
Importance of quality is understood.		

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Terms & Conditions

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Excerpt: SAQA Unit Standard: 116158 - NQF Level 1

Title: Apply basic agricultural enterprise selection principles
Field: Agriculture and Nature Conservation **Sub-field:** Primary Agriculture
US No: 116158 **NQF Level:** 1 **Credits:** 2

Purpose of the Unit Standard:

Qualifying learners will be able to demonstrate an understanding the basic principles of enterprise selection. In addition they will be well positioned to extend their learning and practice into crop production and animal production systems.

This training will benefit the profession by equipping learners with adequate skills to have input into enterprise selection and production to improve productivity and performance.

Learners will understand the importance of the application of business principles in agricultural production with specific reference to enterprise planning.

They will be able to operate farming practices as businesses and will gain the knowledge and skills to move from a subsistence orientation to an economic orientation in agriculture. Farmers will gain the knowledge and skills to access mainstream agriculture through a business-orientated approach to agriculture.

Learning Assumed to be in Place

No learning assumed to be in place.

Unit Standard Range:

Whilst range statements have been defined generically to include as wide a set of alternatives as possible, all range statements should be interpreted within the specific context of application.

Range statements are neither comprehensive nor necessarily appropriate to all contexts. Alternatives must however be comparable in scope and complexity. These are only as a general guide to scope and complexity of what is required.

Specific Outcome (SO) 1:

Name natural resources required for the selection of the relevant enterprise.

Outcome Range: Natural resources include soil, water, climate, vegetation and topography.

Assessment Criteria (AC):

1. Soils are examined and suitability for cultivation is assessed.
2. Water sources are identified.
3. Climatic conditions are identified and described.
4. Basic vegetation types are identified.
5. The topography of the site is recognised and described.

Specific Outcome (SO) 2:

Describe infrastructure requirements for the selection of the relevant enterprise.

Outcome Range: Infrastructure requirements include fencing, housing, water supply, electricity, animal handling facilities and access.

Assessment Criteria (AC):

1. Types of infrastructure are described.
2. The role and function of infrastructure is understood.
3. The availability of infrastructure is determined.
4. The suitability of landscape for infrastructure is determined.

Specific Outcome (SO) 3:

Identify appropriate crops and/or animals for the relevant enterprise.

Outcome Range: All livestock or crops on the farm or garden within a community.

Assessment Criteria (AC):

1. Different livestock or crop types are described.
2. Characteristics of the different types are explained.
3. Different uses of the different types are identified.
4. The suitability of infrastructure for livestock or crops is determined.

Specific Outcome (SO) 4:

Identify production cycle within relevant enterprise.

Outcome Range: All enterprises within the boundaries of the farm within the community

Assessment Criteria (AC):

1. Characteristics of a production cycle are described.
2. The different production cycles are compared.
3. The appropriate production cycle is described correctly.
4. Implementation of the production cycle is observed and reported on.

Specific Outcome (SO) 5:

Identify harvest practice within the relevant enterprise.

Outcome Range: All enterprises within the boundaries of the farm within the community.

Assessment Criteria (AC):

1. A characteristic of harvesting practices is described.
2. Harvesting practices are understood.
3. Importance of health and hygiene is understood.
4. Importance of quality is understood.

Specific Outcome (SO) 6:

Identify post-harvest practice within relevant enterprise.

Outcome Range: All enterprises within the boundaries of the farm within the community.

Assessment Criteria (AC):

1. Characteristics of post harvest practices are described.
2. Post-harvesting practices are understood.
3. Importance of health and hygiene is understood.
4. Importance of quality is understood.

Unit Standard Essential Embedded Knowledge:

The person is able to demonstrate a basic knowledge of:
The role of natural resources.

- The role of Infrastructure.
- The role and importance of Stock.
- The importance of Production cycles.
- The importance of Harvesting practices.
- The role and importance of Post harvest practices.
- Basic farm practices.
- Basic management skills.

Critical Cross-field Outcomes (CCFO):

Working: Teamwork: relates to all specific outcomes;

Collecting: Interpreting Information: relates to all specific outcomes.

Communicating: Communication: relates to all specific outcomes;

Science and Technology: Relates to all specific outcomes;

Demonstrating: Professional development: relates to all specific outcomes;

Self-development: Relates to all specific outcomes.